

Who Pays After Medicaid Expansion?

Coverage, spending and utilisation across payers in US states, 2010–2018

Emily K. Johnson

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The research question

In 2014, the Affordable Care Act offered states the option of expanding Medicaid coverage and opened the Marketplace for private insurance. As a result, total Medicaid spending in expansion states increased as newly eligibles were covered.

We investigate what happened to enrollment, spending and utilization during Medicaid expansion for Medicaid, Medicare, and private insurance across states.

Data and design

IHME Disease Expenditure Project (DEX 2.0): state-level spending and encounters by payer and setting, all ages, 2010–2018. Uninsurance from CPS. **44 states, 396 state-years** — the **25** that expanded on 1 January 2014 and the **19** that never expanded. Later cohorts are dropped ($n \leq 3$ per year).

Callaway and Sant’Anna (2021) group-time difference-in-differences. With one treated cohort, the estimand is the effect in year t on expansion states ($D = 1$):

$$ATT(t) = E[Y_t(1) - Y_t(0) \mid D = 1]$$

$Y_t(0)$ is unobserved after 2013. Writing $\Delta Y_t = Y_t - Y_{2013}$, it is imputed from the never-expanded states by regressing that change on covariates X — poverty rate, median household income, pre-expansion uninsurance, and the shares under 18 and over 65:

$$ATT(t) = E[\Delta Y_t \mid D = 1] - E[m_t(X) \mid D = 1]$$

$$m_t(X) = E[\Delta Y_t \mid X, D = 0]$$

The event-study figures plot $ATT(t)$; the tables report its post-period average.

$$\overline{ATT} = \frac{1}{5} \sum_{t=2014}^{2018} ATT(t)$$

Inference is a multiplier bootstrap, 1,000 draws, clustered on state. **Parallel trends** is a Wald test that $ATT(t) = 0$ in the placebo years 2011–2013. **Multiplicity** is Benjamini–Hochberg across the 36 spending estimates.

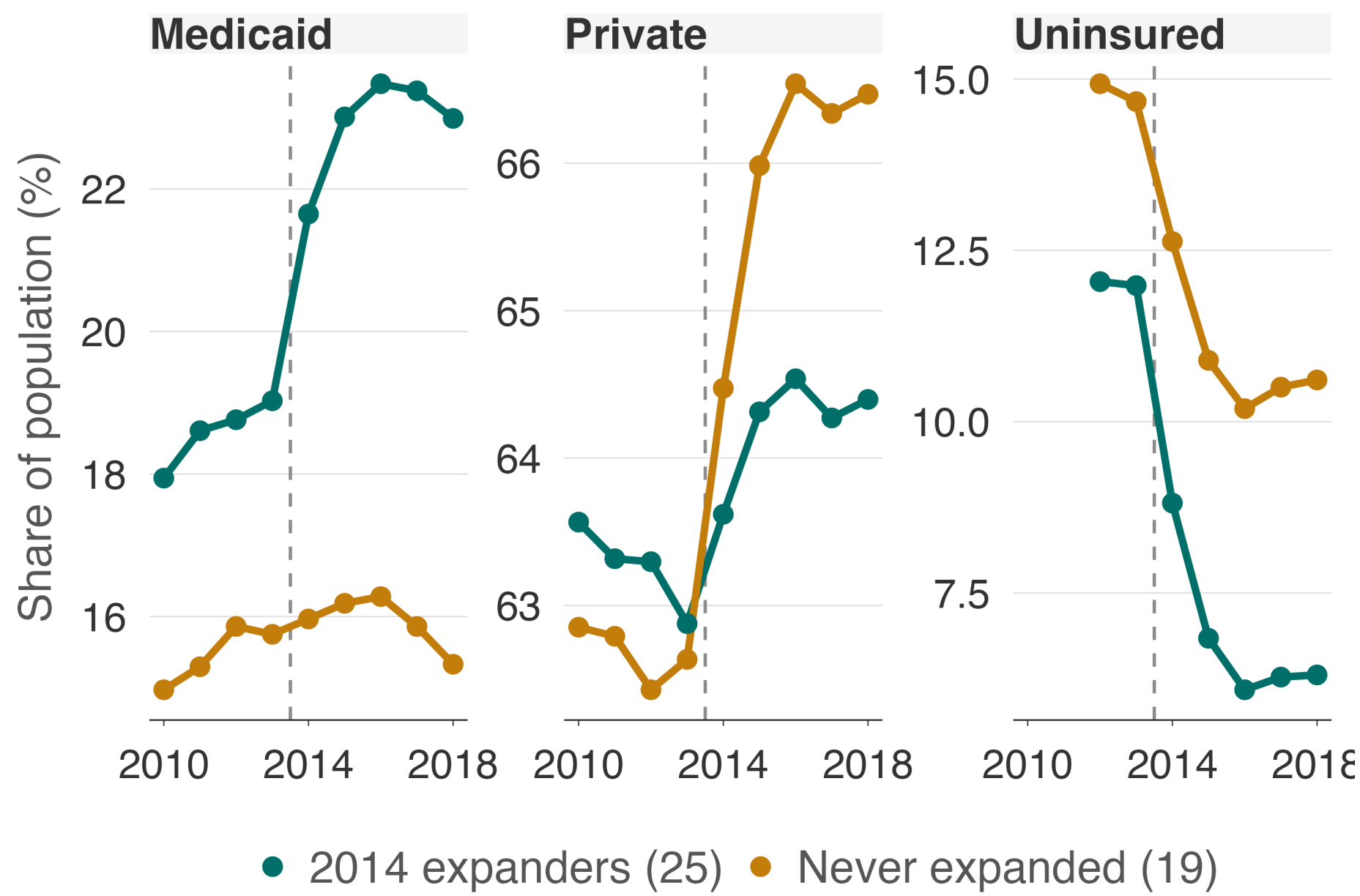
Limitations

Estimates describe 2014 expanders and non-expanders only.

Out-of-pocket spending is not included in analysis. Uninsured spending and out-of-pocket deductibles or copays for the insured cannot be distinguished in DEX.

Private coverage changes cannot be attributed to individual switching.

1. Expansion covered the uninsured

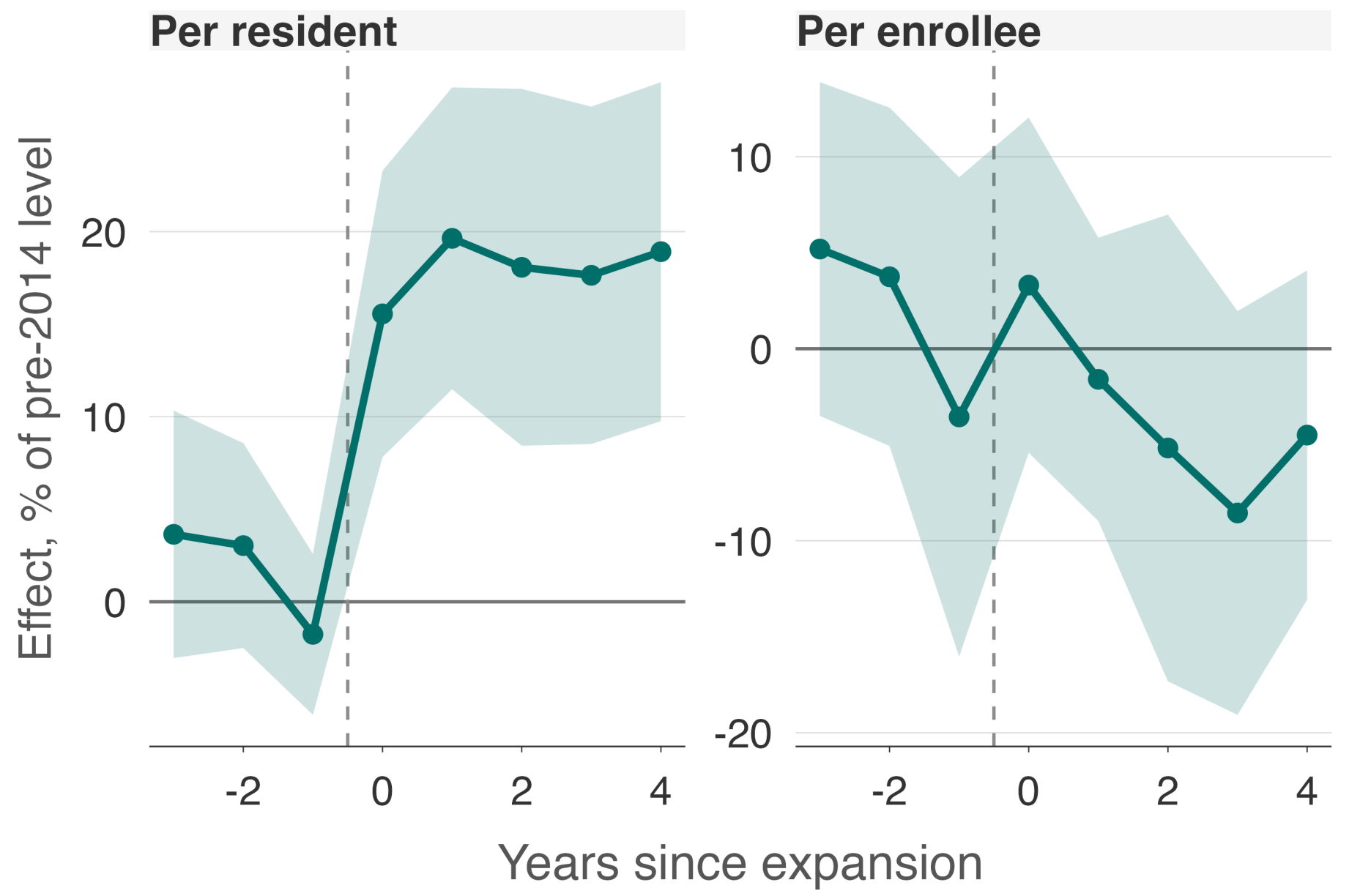


Teal: the 25 states that expanded in 2014. Amber: the 19 that never expanded.

Coverage	Effect	95% CI	Pre-test
Medicaid	+3.69	[+2.27, +5.12]	0.84
Medicare	−0.20	[−0.54, +0.14]	0.72
Uninsured	−1.38	[−2.48, −0.28]	0.30
Private	−2.16	[−3.07, −1.24]	0.96

Medicaid coverage rose **3.7 points** more in expansion states. The uninsured rate fell **1.4 points** further than in control states. Private insurance increased with the Marketplace in both settings, to a greater extent in non-expansion states.

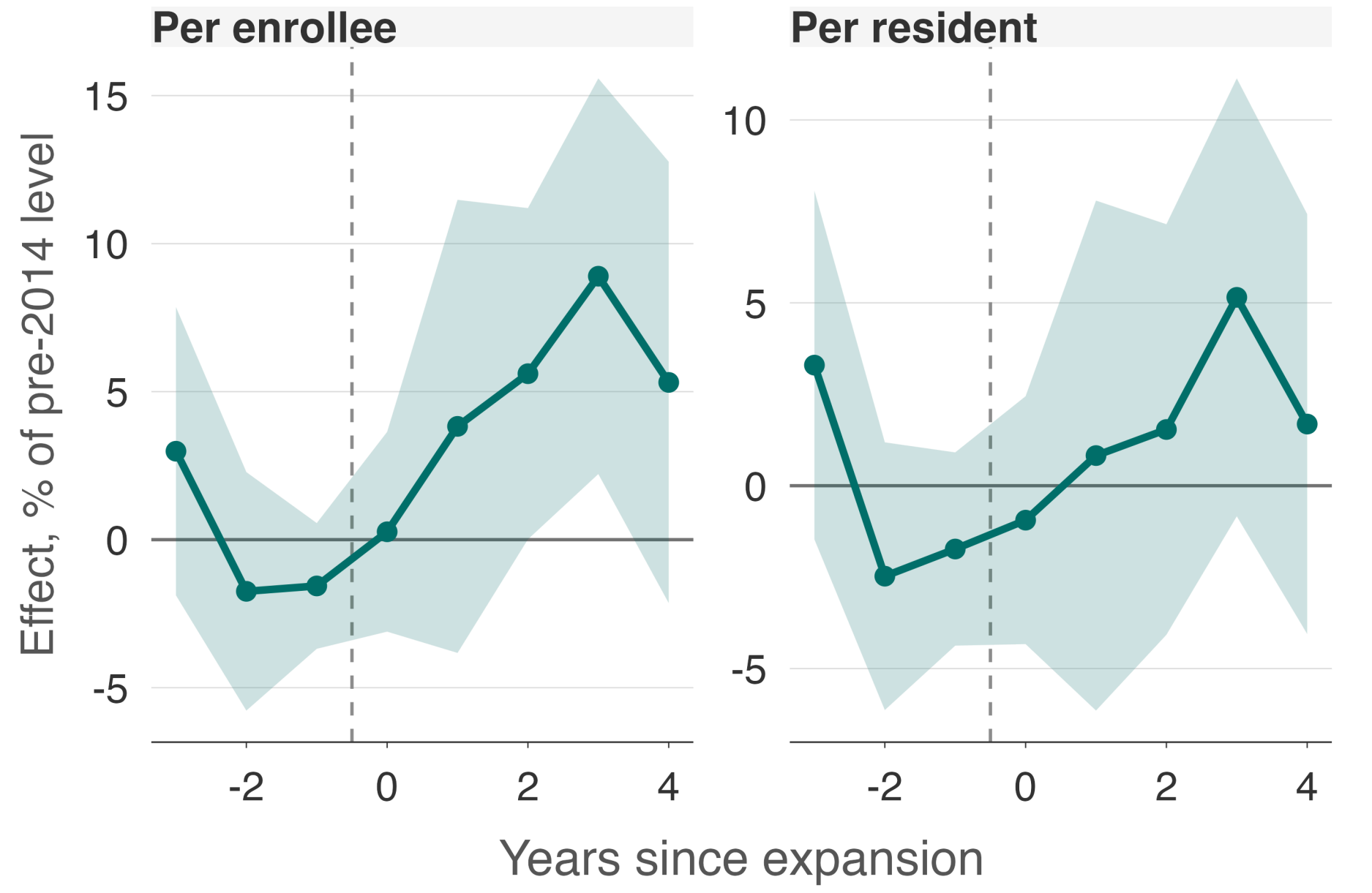
2. New Medicaid enrollees had similar healthcare consumption



Difference-in-differences effect at each year relative to 2014, as a percentage of the pre-expansion level in expansion states. 95% intervals from multiplier bootstrap. Enrollment rose **19.9%** and spending per resident rose **18.0%** (SE 3.9). Spending per enrollee did not follow: **−3.3%** (SE 4.4).

Against a pre-expansion **\$5,649** per enrollee, the marginal enrollee cost roughly **\$4,500** per year.

3. Weak evidence of increased private spending



Private coverage rose in both groups — 62.9 to 64.4% in expanders, 62.6 to 66.5% in controls. The **−2.2 point** estimate is control states gaining faster, not expansion states losing. Marketplace subsidies cover 100–138% FPL where Medicaid does not.

Private spending per enrollee **+4.8%** (SE 2.8) and per resident **+1.7%** (SE 2.4). Neither reaches significance. Spending did not fall on either margin.

Encounters per enrollee are positive in all five settings (**+2.2 to +5.6%**) and significant in none, so what movement there is looks like **use rather than price**. The most likely reading is dilution of the *control* group’s risk pool as lower-utilising marketplace enrollees joined it.

Axes as in panel 2. Encounters are not summed across settings: a prescription fill and a hospital stay are not the same unit.

4. Nothing else moved

Medicare spending per resident **+0.4%** (SE 1.3) — nothing, on any margin. Private spending per resident **+1.7%** (SE 2.4).

Of the **36 spending estimates** below, three survive correction for multiple comparisons, and all three are Medicaid spending per resident. Among the **24 non-Medicaid** estimates, one clears 5% before correction and **none survives it**.

Spending per resident

Setting	Medicaid	Medicare	Private
All care	+18.0 (3.9) *	+0.4 (1.3)	+1.7 (2.4)
Ambulatory	+23.6 (5.8) *	−0.9 (1.2)	+2.1 (2.4)
Emergency	+21.7 (8.1)	+2.9 (4.9)	−0.2 (1.9)
Inpatient	+22.5 (4.5) *	+0.5 (1.4)	+1.2 (3.4)
Nursing	+2.0 (6.9)	+3.0 (5.6)	+4.6 (7.2)
Pharmacy	+11.1 (12.8)	+0.8 (2.9)	+0.8 (5.2)

Spending per enrollee

Setting	Medicaid	Medicare	Private
All care	−3.3 (4.4)	+1.5 (0.8)	+4.8 (2.8)
Ambulatory	+0.0 (6.2)	+0.9 (0.8)	+5.3 (3.0)
Emergency	−2.0 (7.8)	+3.8 (2.7)	+3.2 (2.2)
Inpatient	+4.4 (4.9)	+0.8 (0.9)	+4.0 (3.6)
Nursing	−14.4 (9.7)	+4.1 (4.9)	+7.3 (7.3)
Pharmacy	−22.5 (18.0)	+3.2 (1.4)	+4.2 (4.7)

Effect as % of the pre-2014 level (SE). * survives Benjamini–Hochberg correction across the 36 estimates. Every Medicaid model passes the parallel-trends pre-test; private is borderline at $p = 0.07$.

Expansion’s new enrollees cost about as much as the old ones, and private insurance did not shrink to pay for them.

Medicaid spending per resident rose 18%; per enrollee it did not move. Private coverage grew in expansion states too — just more slowly than where the marketplace did the covering.